

Dennis G. Pezan

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EDUCATION

United States Military Academy at West Point

West Point, NY

B.S., Operations Research – Finance Specialization, AI Minor

Class of 2029

- GPA: 3.586 | Dean's List, 26-1 and 26-2
- **Stokes Writing Fellowship** – endowed fellowship for selected cadet writers; funds research travel, expert mentorship, and publication of original research

EXPERIENCE

MIT Center for Transportation & Logistics – USMA Research

West Point, NY

Undergraduate Researcher, Network Interdiction for Contested Logistics

2026–Present

- Investigate how to optimally attack and defend military logistics networks under adversarial interdiction, applying network optimization, game theory, and Markov decision processes

United States Military Academy – AI Research

West Point, NY

Undergraduate Research Assistant, Department of Mathematical Sciences

2025–Present

- Conduct applied AI and ML research; contribute to projects exploring model behavior and data-driven decision making
- Developed Python pipelines (PyTorch, scikit-learn) to train and benchmark reinforcement learning and ensemble models; automated experiment tracking and visualization to accelerate research iteration

Transformation Decision Analysis Center (TDAC), U.S. Army

Fort Lee, VA

Research Intern, Modeling & Analysis Division

Summer 2026

- Engineered and iteratively redesigned the reward function for a reinforcement learning agent in the Army's Logistics Battle Command (LBC) simulation, reshaping how the agent prioritized logistics decisions
- Trained and evaluated the agent in Python (PyTorch) against a new implementation of the LBC environment, analyzing how reward design shaped learned behavior

McDonald's

Plainfield, IL

Department Manager

September 2022 – July 2024

- Managed weekly scheduling for 80+ employees with 95% on-time accuracy; oversaw \$20,000+ in weekly supply procurement and operations
- Promoted as the youngest department manager in the region; sustained 90%+ customer satisfaction

PROJECTS

Tree-Based Methods Research Site

dennis-pezan.com/ma289.html

- Built and deployed an interactive site analyzing Decision Trees, Bagging, and Random Forests on the Titanic dataset in Python (scikit-learn); trained and evaluated all three on an identical 80/20 split
- Computed pairwise tree correlation, overfitting gaps, and feature importance to show how ensemble decorrelation reduces variance; visualized with matplotlib and deployed via GitHub and Vercel

CFT Packing List – Web Application

dennis-pezan.com/cft.html

- Developed and deployed a Python web tool helping cadets organize and track gear for Cadet Field Training; hosted on Vercel with automatic deployment from GitHub

LEADERSHIP & ACTIVITIES

CLD-STEM / Cadet Ambassador Program

West Point, NY | 2025–Present

- Represent West Point at advisory boards in Chicago and Los Angeles
- Delivered STEM outreach and mentorship to underrepresented students (50+ hours)

Center for Enhanced Performance (CEP) | Cadet in Charge, MA104 Tutoring

West Point, NY | 2026–Present

- Lead the full tutor staff for MA104, a core mathematics course with 1,010 cadets enrolled; direct tutoring coverage for every major graded event across the academic year
- Serve as a certified CEP tutor in math, chemistry, and psychology

SKILLS

Technical: Python (PyTorch, scikit-learn), Java (Spring Boot), SQL, Excel, Microsoft Fabric, Kafka, REST APIs

Languages: Spanish (fluent)